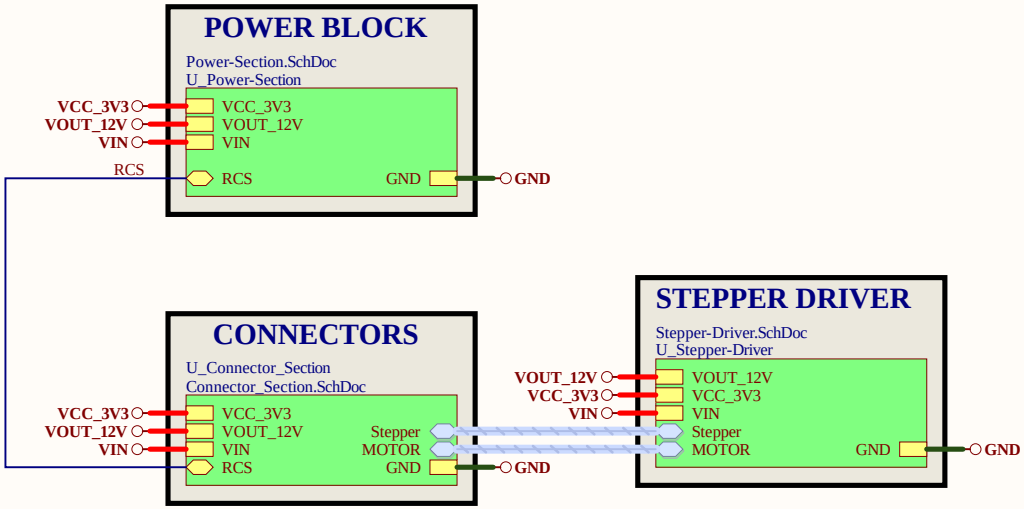
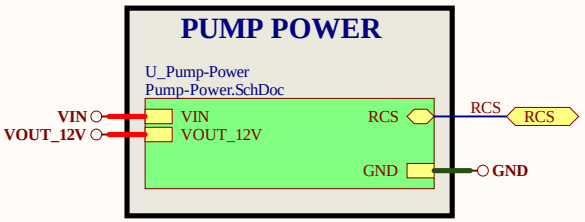
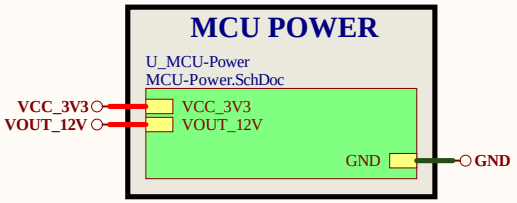
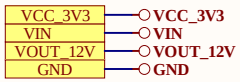
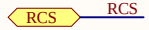

POWER BLOCK
Power-Section.SchDoc
U_Power-Section
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
Stepper
MOTOR
GND
GND
STEPPER DRIVER
Stepper-Driver.SchDoc
U_Stepper-Driver
VOUT_12V
VCC_3V3
VIN
VOUT_12V
VCC_3V3
VIN
Stepper
MOTOR
GND
GND


PUMP POWER
U_Pump-Power
Pump-Power.SchDoc
VIN
VOUT_12V
VIN
VOUT_12V
RCS
GND
RCS
RCS
GND
GND

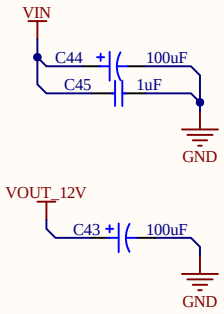

MCU POWER
U_MCU-Power
MCU-Power.SchDoc
VCC_3V3
VOUT_12V
VCC_3V3
VOUT_12V
GND
GND


VCC_3V3
VIN
VOUT_12V
GND
VCC_3V3
VIN
VOUT_12V
GND

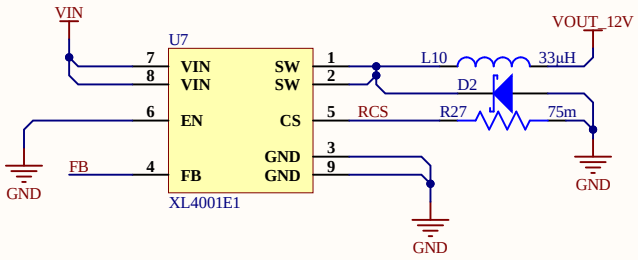
CURRENT SENSE INPUT



DECOUPLING CAPACITORS



12V BUCK REGULATOR

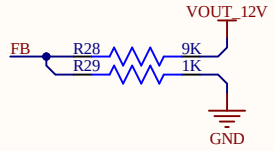


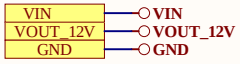
VIN=36V
VOUT=12V, where VOUT= 1.235(1+R2/R1)*
For 12V output, R2=9K and R1=1K
(Datasheet pg.3)

For IOUT=2A, RCS=75mA, (RCS=0.155/IOUT)

Reference: XL4001 Datasheet

FEEDBACK NETWORK

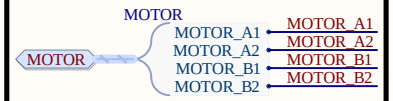




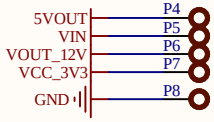
CONTROL SIGNALS



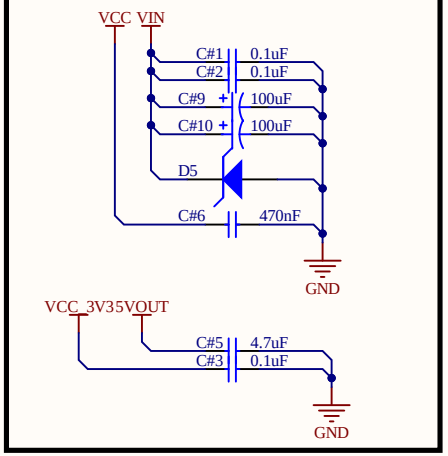
MOTOR OUT



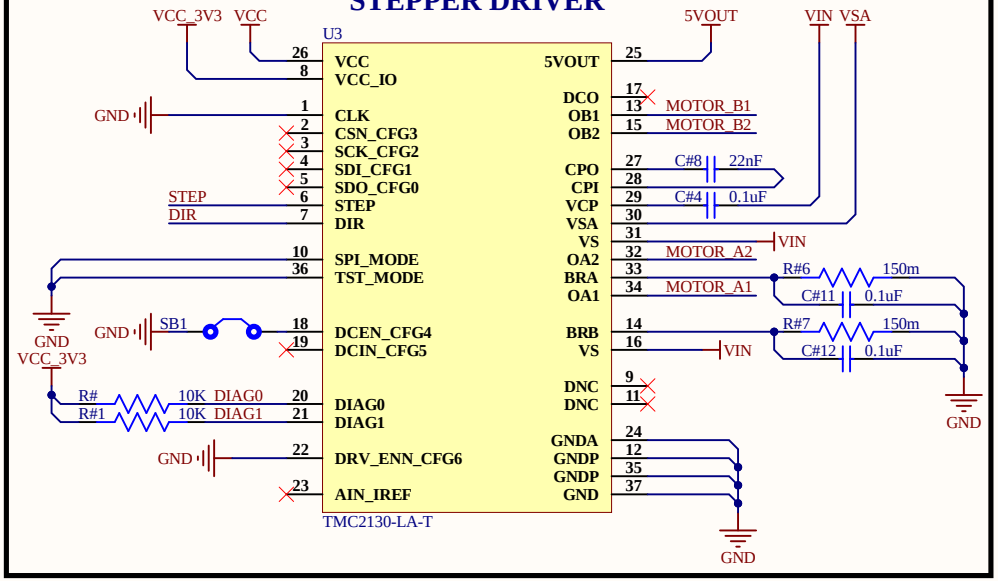
TEST POINTS



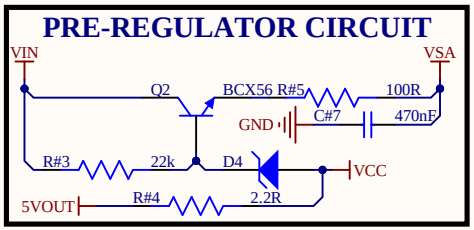
DECOUPLING CAPACITORS



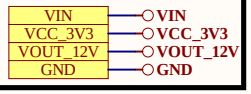
STEPPER DRIVER



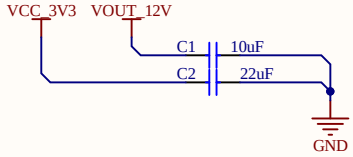
PRE-REGULATOR CIRCUIT



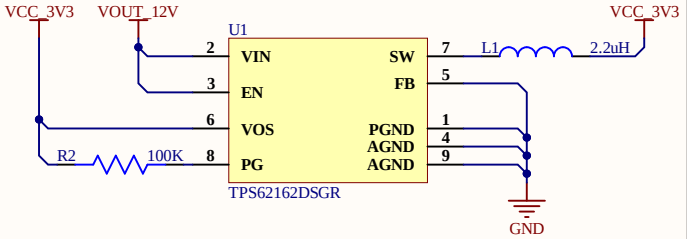
POWER SIGNALS



DECOUPLING CAPACITORS

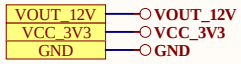


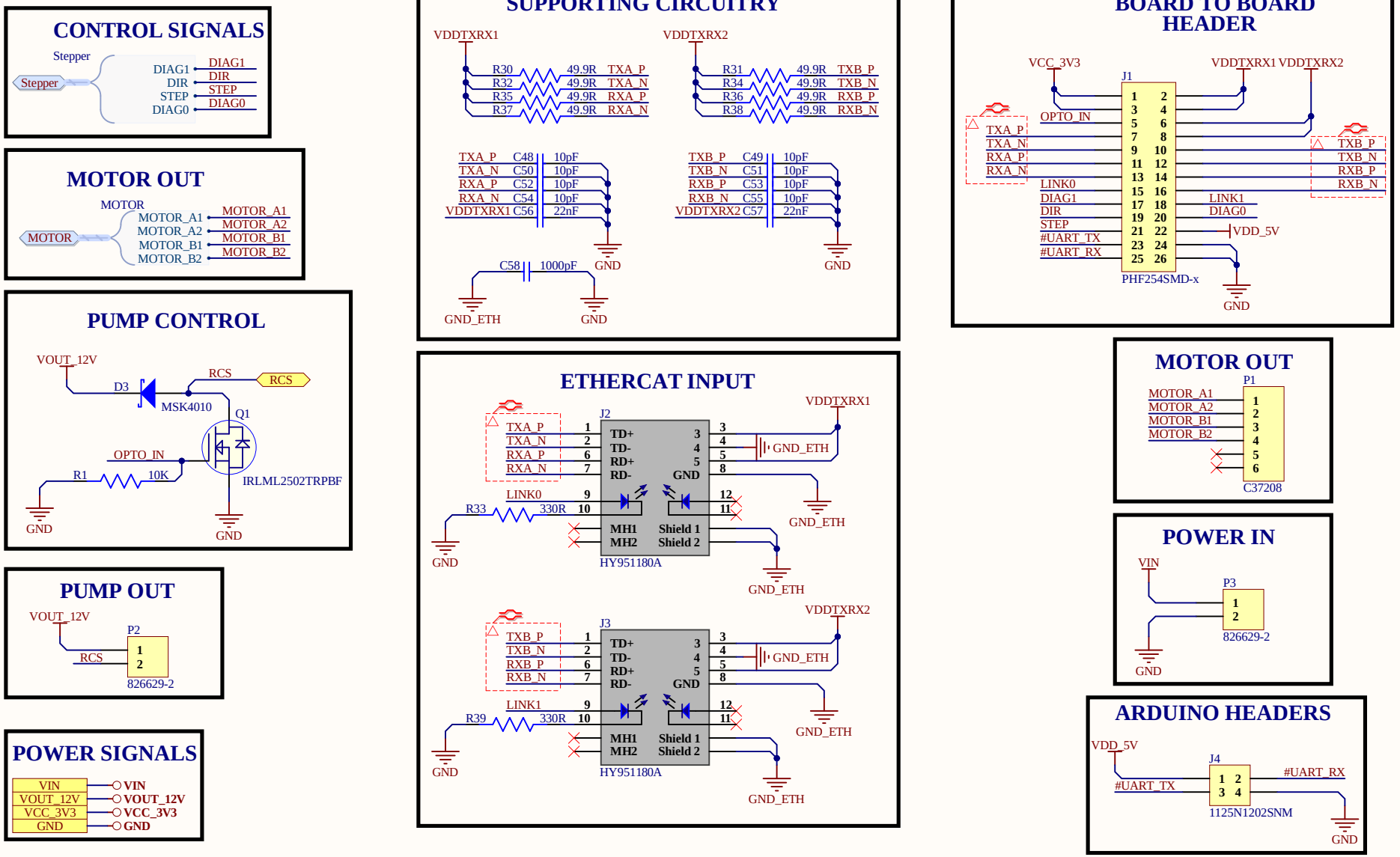
3V3 BUCK REGULATOR



Fixed regulator.
VOUT= 3V3/1A

POWER SIGNALS




CONTROL SIGNALS
Stepper
DIAG1
DIR
STEP
DIAG0
MOTOR OUT
MOTOR
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
PUMP CONTROL
VOUT_12V
D3
MSK4010
RCS
Q1
OPTO_IN
R1
10K
IRLML2502TRPBF
GND
PUMP OUT
P2
826629-2
POWER SIGNALS
VIN
VOUT_12V
VCC_3V3
SUPPORTING CIRCUITRY
VDDTXRX1
VDDTXRX2
R30 49.9R TXA_P
R32 49.9R TXA_N
R35 49.9R RXA_P
R37 49.9R RXA_N
R31 49.9R TXB_P
R34 49.9R TXB_N
R36 49.9R RXB_P
R38 49.9R RXB_N
C48 10pF
C50 10pF
C52 10pF
C54 10pF
C56 22nF
C49 10pF
C51 10pF
C53 10pF
C55 10pF
C57 22nF
C58 1000pF
GND_ETH
ETHERCAT INPUT
J2
J3
TD+
TD-
RD+
RD-
LINK0
LINK1
R33 330R
R39 330R
MH1
MH2
Shield 1
Shield 2
HY951180A
BOARD TO BOARD HEADER
VCC_3V3
J1
OPTO_IN
LINK0
LINK1
DIAG1
DIAG0
DIR
STEP
#UART_TX
#UART_RX
VDD_5V
PHF254SMD-x
MOTOR OUT
P1
C37208
POWER IN
P3
826629-2
ARDUINO HEADERS
J4
1125N1202SNM

Comment	Description	Designator	Footprint	LibRef	Quantity
10uF	CAP CER 10UF 25V X5R 0603	C1	Cap0603	GRM188R61E106MA73J	1
22uF	CAP CER 22UF 25V X5R 0805	C2	Cap0805	CC0805MKX5R8BB226	1
100uF	25V 5mm 10mm 100uF 100kHz SMD,5x10mm Aluminum Electrolytic Cap- SMD ROHS	C43	VKMB1001E101MV	VKMB1001E101MV	1
100uF	63V 10mm 8mm 100uF ±20% ,8x10mm Aluminum Electrolytic Cap	C44, C#9, C#10	VKMD1001J101MV	VKMD1001J101MV	3
1uF	CAP CER 1UF 100V X7S 0805	C45	Cap0805	GCM21BC72A105KE36L	1
10pF	CAP CER 10PF 50V C0G/NP0 0603	C48, C49, C50, C51, C52, C53, C54, C55	Cap0603	06035A100JAT2A	8
22nF	CAP CER 0.022UF 100V X7R 0603	C56, C57, C#8	Cap0603	CGA3E2X7R2A223K080AA	3
1000pF	1206 1 nF 2 kV ±10 % Tolerance X7R SMT Multilayer Ceramic Capacitor	C58	Cap1206	CC1206KKX7RDBB102	1
0.1uF	CAP CER 0.1UF 100V X7R 0603	C#1, C#2, C#3, C#4, C#11, C#12	Cap0603	CC0603KRX7R0BB104	6
4.7uF	CAP CER 4.7UF 25V X5R 0603	C#5	Cap0603	C1608X5R1E475M080AC	1
470nF	CAP CER 0.47UF 100V X7S 0805	C#6, C#7	Cap0805	C2012X7S2A474KT000N	2
PMEG3030EP,115	Diode Schottky 30V 3A 2-Pin SOD-128 T/R	D2	PMEG3030EP,115	PMEG3030EP,115	1
MSK4010	SOD-523 Schottky Barrier Diodes (SBD) ROHS	D3	MSK4010	MSK4010	1
MMSZ5V6CW	1μA@2V 40Ω Single 5.2V~6V 500mW 5.6V SOD123G Zener Diodes ROHS	D4	SOD123G	MMSZ5V6CW	1
SMBJ43A	8.6A 69.4V 47.8V 43V Unidirectional SMB(DO-214AA) TVS ROHS	D5	SMB	SMBJ43A	1
PHF254SMD-x	P=2.54mm Female Headers,SMD,ROHS	J1	PHF254SMD-x	PHF254SMD-x	1
HY951180A	Through Hole Ethernet Connectors/Modular Connectors (RJ45 RJ11) ROHS	J2, J3	HY951180A	HY951180A	2
1125N1202SNM	CONN HEADER 4 POS 2.54mm SMD	J4	1125N1202SNM	1125N1202SNM	1
2.2uH	IND SMD 1.5A 2.2uH ±20% 135mΩ	L1	MCS20FC-2R2MMP	MCS20FC-2R2MMP	1
33μH	FIXED IND 33UH 2.5A 189.2MOHM SM	L10	ETQ-P4M330YFN	ETQ-P4M330YFN	1
C37208	Straight 1 6 2.54mm Black Brass Plugin,P=2.54mm Pin Headers ROHS	P1	C37208	C37208	1
826629-2	CONN HEADER VERT 2POS 2.54MM	P2, P3	HDR1X2	826629-2	2
Test Point	Test Point ON	P4, P5, P6, P7, P8	TEST POINT	TEST POINT	5
IRLML2502TRPBF	MOSFET N-CH 20V 4.2A SOT23	Q1	IRLML2502TRPBF	IRLML2502TRPBF	1
BCX56	80V 500mW 1A 63@150mA,2V 130MHz 500mV@500mA,50mA NPN +150°C@(Tj) SOT-89(SOT-89-3) Bipolar Transistors - BJT ROHS	Q2	SOT-89	BCX56	1
10K	RES SMD 10K OHM 5% 1/10W 0603	R1, R#, R#1	Res0603	AC0603JR-0710KL	3
100K	RES SMD 100K OHM 5% 1/5W 0603	R2	Res0603	CRGH0603J100K	1
75m	RES 0.075 OHM 1% 2W 2512	R27	RES 2512	CSRN2512FK75L0	1
9K	RES SMD 9kΩ ±100ppm/°C 0.125W 0805	R28	RES 0805	0805W8F9001T5E	1
1K	RES SMD 1K OHM 1% 1/10W 0603	R29	Res0603	RC0603FR-071KL	1
49.9R	RES 49.9 OHM 1% 1/10W 0603	R30, R31, R32, R34, R35, R36, R37, R38	Res0603	RC0603FR-0749R9L	8
330R	RES SMD 330R 100 mW 0603	R33, R39	Res0603	RC0603JR-07330RL	2
22k	RES SMD 22K OHM 0.1% 1/10W 0603	R#3	Res0603	RT0603BRE0722KL	1
2.2R	RES 2.2 OHM 1% 1/10W 0603	R#4	Res0603	RC0603FR-072R2L	1
100R	RES SMD 100 OHM 1% 1/10W 0603	R#5	Res0603	RC0603FR-07100RL	1
150m	RES SMD 150m ohms 1/4 1206	R#6, R#7	Res-1206	1206W4F150LT5E	2
Solder Bridge	Solder Bridge ON	SB1	Solder Bridge	Solder Bridge	1
TPS62162DSGR	Buck Switching Regulator IC Positive Fixed 3.3V 1 Output 1A 8-WFDFN Exposed Pad	U1	8-WSON (2x2)	TPS62162DSGR	1
TMC2130-LA-T	IC STEPPER MOTOR DVR SPI 36QFN	U3	TMC2130-LA-T	TMC2130-LA-T	1
XL4001E1	2A 40V SOIC-8_EP_150mil DC-DC Converters ROHS	U7	XL4001E1	XL4001E1	1